



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Data Science Lab (DSL) — Course Project Report

Credit Risk Modelling:

An Explainable Credit Scoring System: Integrating Default Probability
Estimation with CIBIL Score Calculation and Borrower Guidance

Under the guidance of: **Swaminathan Rammohan**

Submitted by:

Hrishav Majumdar	25BM6JP15	
Kingshuk Barua	25BM6JP23	
Patil Rohit Ranjit	25BM6JP34	April 2026
Pritam Sarkar	25BM6JP40	
Siddharth Sharan	25BM6JP48	

Abstract

This report presents an ML-driven credit risk modelling framework for Indian banks and NBFCs. Using the “Give Me Some Credit” dataset (150K borrower records, 11 features), we train Logistic Regression, XGBoost, CatBoost, and a Mixture-of-Experts (MoE) neural network. CatBoost, optimised via Optuna (50 TPE trials), achieves ROC-AUC ≈ 0.86 . A dual-threshold framework (Max-F1 and Precision-targeted) enables calibrated precision–recall trade-offs. SHAP provides per-feature explanations with actionable recommendations. A CIBIL-aligned score (300–900) weights default probability at 50%, payment history 25%, credit utilisation 15%, credit mix 10%. Deployed via Next.js CRM dashboard and Google Sheets API.

Summary of Contributions

- Thorough EDA through univariate, bivariate plots, and correlation analysis using statistical methods: Point-biserial r , Mann-Whitney U , Rank-biserial r , Cohen’s d , AUC (continuous vs binary) and χ^2 , Cramér’s V , Odds Ratio, Cochran-Armitage trend, Somers’ D (discrete vs binary).
- Built a production-grade credit risk engine combining gradient boosting (CatBoost) and neural MoE models plus distillation, achieving ~ 0.86 ROC-AUC and enabling probabilistic risk scoring.
- Developed and benchmarked 4 model architectures (Logistic Regression, XGBoost, CatBoost, Mixture-of-Experts), improving performance over baseline by $\sim 4\%$ AUC.
- Designed a dual-threshold decision framework (Max-F1 & Precision-constrained) enabling flexible risk calibration for different business scenarios.
- Identified key business insights such as delinquency features as dominant predictors and importance of threshold selection over model choice.
- Built a production-ready ML pipeline with modular architecture (data preprocessing, training, evaluation, deployment) ensuring scalability and reproducibility.
- Integrated model explainability (SHAP) into predictions, translating black-box outputs into actionable borrower-level insights.
- Deployed an end-to-end ML system (Python + React) enabling real-time inference and scalable batch processing.

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1 Introduction

Credit risk assessment is critical in Indian financial services (INR 150 lakh crore disbursed by FY 2024–25; gross NPAs $\approx 2.8\%$). The current landscape suffers from:

- **Static scorecards** that miss non-linear interactions.
- **Opaque decisions** without actionable borrower explanations.
- **Reactive posture** — point-in-time rather than predictive.
- **Threshold rigidity** — single boundary instead of calibrated trade-offs.

This project develops an ML-driven framework combining predictive accuracy with SHAP-based explainability, CIBIL-aligned scoring, and real-time deployment.

2 Dataset Description

The project uses the “Give Me Some Credit” dataset containing 150,000 training observations with 11 financial attributes.

Table 1: Feature descriptions

Kaggle Name	Internal Name	Type	Description
SeriousDlqin2yrs	defaulted	Binary	Target: 90+ days past due in 2 yr
RevolvingUtilization...	unsecured_credit	Float	Revolving utilisation ratio
age	age	Integer	Borrower age in years
NumberOfTime30–59...	delinq_30_59	Count	30–59 day late payments
DebtRatio	debt_ratio	Float	Monthly debt / income
MonthlyIncome	monthly_income	Float	Monthly salary (USD)
NumberOfOpenCredit...	open_credit	Count	Open credit lines
NumberOfTimes90DaysLate	delinq_90	Count	90+ day late events
NumberRealEstate...	real_estate_loans	Count	Home/mortgage loans
NumberOfTime60–89...	delinq_60_89	Count	60–89 day late payments
NumberOfDependents	dependents	Count	Household dependents

Key statistics: 150,000 rows $\rightarrow$ **112,021** after cleaning. Default rate: 6.74% ($\approx 13.5:1$ class imbalance). Values 96 and 98 in delinquency fields are sentinel codes (not actual counts).

3 Data Cleaning & Preprocessing Pipeline

Shared across all notebooks:

1. **Column rename:** Kaggle names $\rightarrow$ snake_case.
2. **Remove age ≤ 0 :** Impossible ages filtered out.
3. **Sentinel flag:** $\text{delinq_90} \in \{96, 98\} \rightarrow \text{delinq_sentinel} = 1$ (54.6% default rate vs 6.6%).
4. **Delinquency cap:** All delinq columns clipped to ≤ 10 .
5. **Missing indicators:** `monthly_income_missing` (19.8% MNAR), `dependents_missing` (2.6% MCAR).
6. **Median imputation:** Income and dependents filled with column median.
7. **Train/Val/Test split:** 70/15/15% stratified.

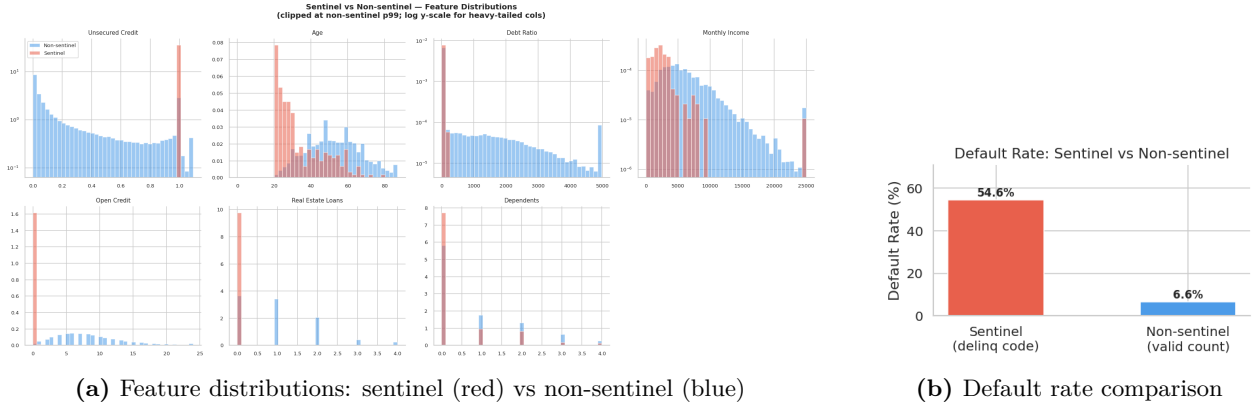
Final: 13 features (10 original + 3 engineered).

4 Exploratory Data Analysis

4.1 Sentinel Accounts & Missingness

Two key data-quality findings emerge before any modelling:

Sentinel codes in delinquency fields. Values 96 and 98 in the delinquency columns are data-entry codes, not actual past-due counts. Only 269 accounts (0.18%) carry these codes, but they exhibit a **54.6% default rate** vs. 6.6% for non-sentinel borrowers — an $8\times$ difference. We retain these rows, flag them with `delinq_sentinel = 1`, and cap all delinquency counts at 10.

**Figure 1:** Sentinel vs non-sentinel account profiles

Sentinel borrowers are younger (median age 33 vs 52), have near-maximal credit utilisation, and carry substantially lower incomes — a high-risk profile consistent with the extreme default rate.

Missing values. Only two columns have missing data:

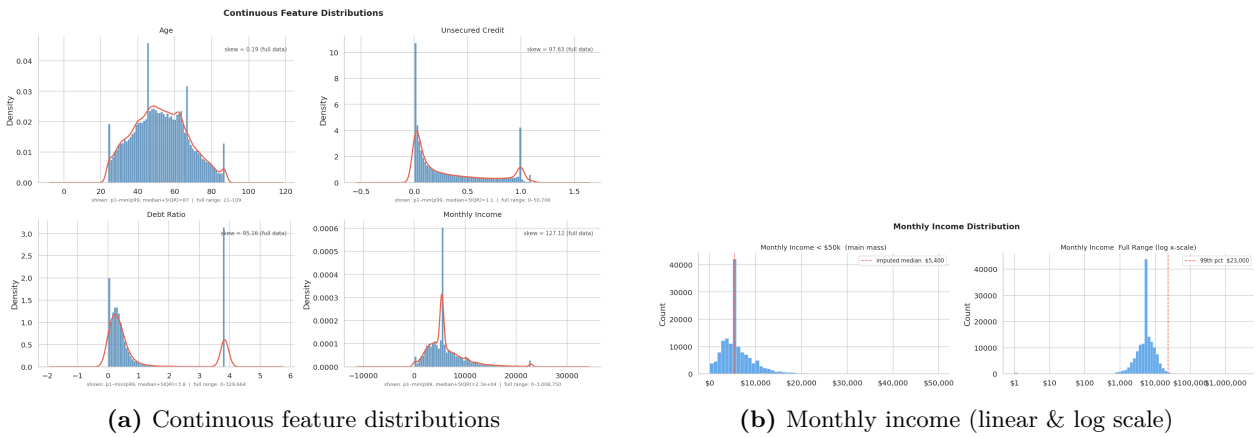
Table 2: Missing value summary

Column	Missing	%	Mechanism
monthly_income	29,731	19.8%	MNAR — missing borrowers default less (5.6% vs 6.9%)
dependents	3,924	2.6%	MCAR — no association with target

The counter-intuitive finding that missing income is *protective* suggests MNAR: financially stable borrowers (self-employed, investment income) are less likely to report salaried income. Binary indicators `monthly_income_missing` and `dependents_missing` are created before median imputation to preserve this signal.

4.2 Univariate Analysis — Continuous Features

Histograms with KDE overlays reveal extreme right-skew in `monthly_income`, `debt_ratio`, and `unsecured_credit`. Log1p transforms are applied for visualisation and for linear/neural models.

**Figure 2:** Continuous feature distributions

4.3 Discrete Feature Distributions

Delinquency variables are heavily zero-inflated (>85% at 0). `open_credit` follows a near-Poisson distribution.

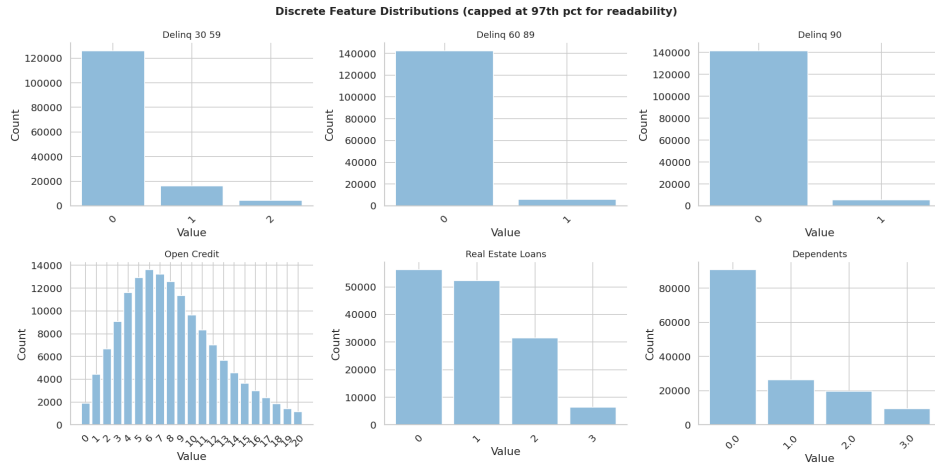


Figure 3: Discrete feature distributions (capped at 99.99th percentile)

4.4 Class Imbalance

The dataset exhibits a **13.5:1** imbalance ratio (6.74% defaulted). Handling: LR uses `class_weight='balanced'`; XGBoost uses `scale_pos_weight ≈ 14`; CatBoost uses `class_weights`; MoE uses `pos_weight = 7.0`.

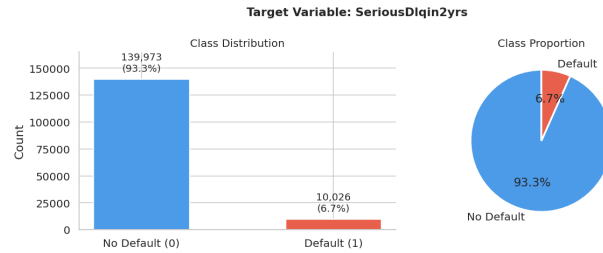


Figure 4: Target class distribution

4.5 Bivariate Analysis — Features vs Default

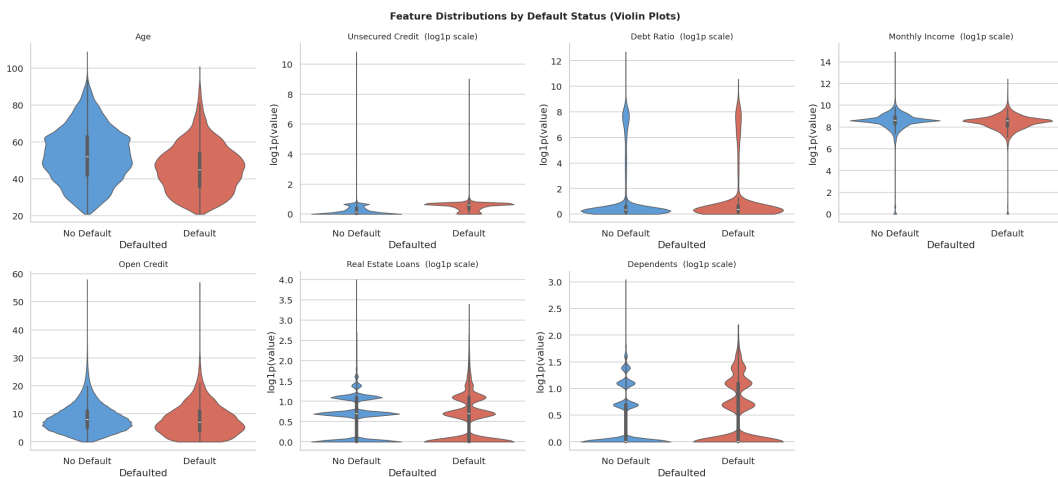
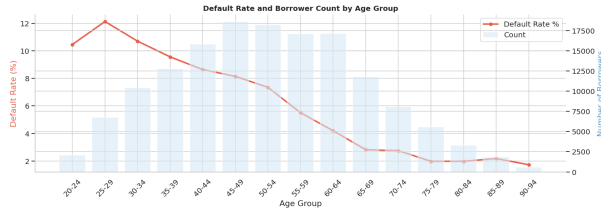
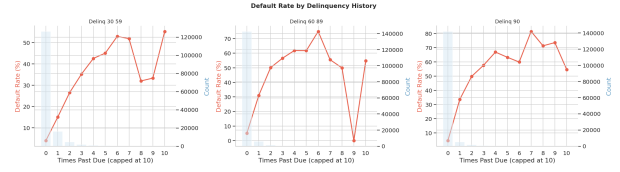


Figure 5: Violin plots — feature distributions split by default status



(a) Default rate by age group



(b) Default rate by delinquency

Figure 6: Bivariate relationships with default

5 Correlation Analysis

With a binary target and mixed input types, different association measures are needed:

Table 3: Appropriate association measures by input type

Input	Target	Measures
Continuous	Binary	Point-biserial r , Mann-Whitney U , Rank-biserial r , Cohen's d , AUC
Discrete	Binary	χ^2 , Cramér's V , OR, Cochran-Armitage, Somers' D
Feature $\times$ Feature	—	VIF

5.1 Continuous Features vs Binary Target

Table 4: Continuous features vs default — association measures. PB r = Point-biserial (linear); Rank-bis. r = effect size for Mann-Whitney U ; AUC = single-feature discriminative power.

Feature	PB r	Rank-bis. r	Cohen's d	AUC
unsecured_credit	−0.002	−0.556	−0.007	0.222
age	−0.115	+0.271	−0.465	0.635
monthly_income	−0.017	+0.142	−0.069	0.571
debt_ratio	−0.008	−0.048	−0.030	0.476

Key findings: `unsecured_credit` has a single-feature AUC of 0.222 (inverted > 0.778) — high utilisation is a strong stress flag. `age` shows the largest rank-biserial r (+0.271, AUC = 0.635). `monthly_income`: Raw Cohen's d (−0.069) is distorted by extreme skew; rank-biserial r (+0.142) is more reliable. `debt_ratio`: Weak linear association (PB r = −0.008) despite conceptual importance.

5.2 Discrete / Count Features vs Binary Target

Table 5: Discrete features vs default — association measures. χ^2 = Chi-square; V = Cramér's V ; Somers' D = directional ordinal association; CA z = Cochran-Armitage trend statistic; OR = odds ratio (binary inputs only).

Feature	χ^2	Cramér's V	Somers' D	CA z	OR [95% CI]
delinq_90	17895.5	0.345	+0.362	+120.7	—
delinq_60_89	11461.4	0.276	+0.305	+102.9	—
delinq_30_59	11429.2	0.276	+0.163	+105.1	—
open_credit	1692.7	0.106	−0.012	−11.5	—
delinq_sentinel	986.2	0.081	+0.481	—	17.06 [13.41, 21.70]
real_estate_loans	740.9	0.070	−0.014	−2.7	—
dependents	354.5	0.049	+0.020	+18.2	—
mi_missing	67.9	0.021	−0.013	—	0.80 [0.75, 0.84]
dep_missing	28.8	0.014	−0.022	—	0.66 [0.57, 0.77]

All three delinquency features show CA $z > 100$ and positive Somers' D , confirming strong monotonic dose-response trends (see Fig. 7). `delinq_sentinel` has OR = 17.06 — the strongest single predictor. `monthly_income_missing` has OR = 0.80 — missing income is *protective* (MNAR mechanism). All p-values survive Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

5.3 Variance Inflation Factor (VIF)

$VIF_j = 1/(1 - R_j^2)$ measures how much a feature's variance is inflated by multicollinearity. Thresholds: $VIF < 5$ (low), $5 \leq VIF < 10$ (moderate), $VIF \geq 10$ (severe).

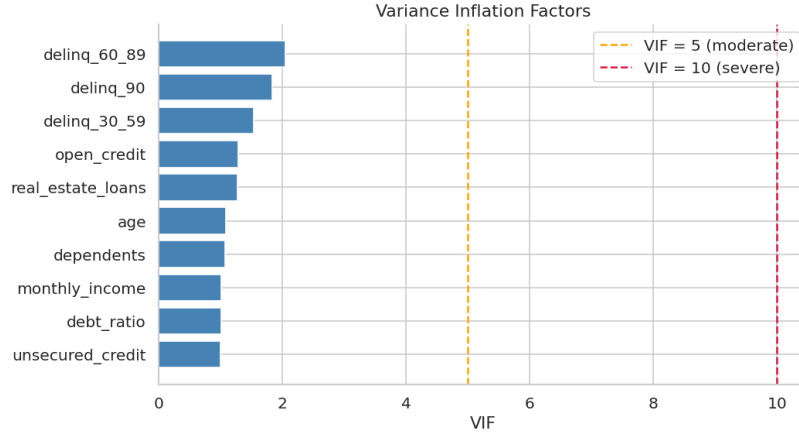


Figure 7: VIF — all features have $VIF < 5$, indicating no problematic multicollinearity.

5.4 Mutual Information

Mutual information captures **any** (including non-linear) dependence: $MI(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$.

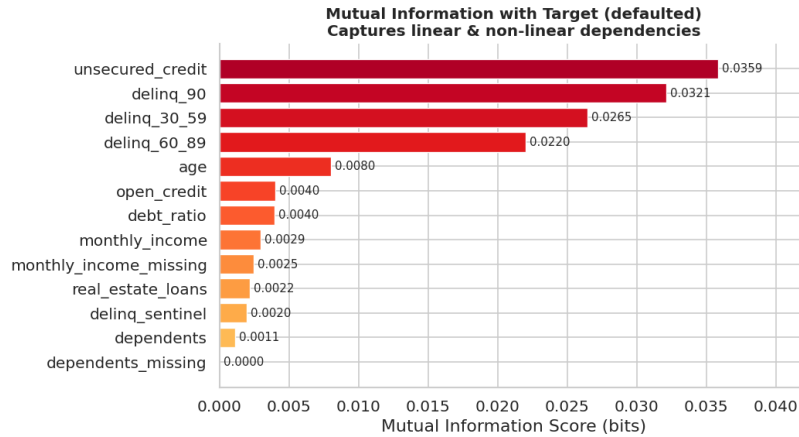


Figure 8: Mutual information ranking (bits)

Top 5: delinq_90 (0.487), delinq_60_89 (0.386), delinq_30_59 (0.384), unsecured_credit (0.198), age (0.156).

6 Modelling

Six models were trained and evaluated on a consistent 70/15/15 stratified split. All models use the same 13-feature set. Two decision thresholds are tuned on the validation set:

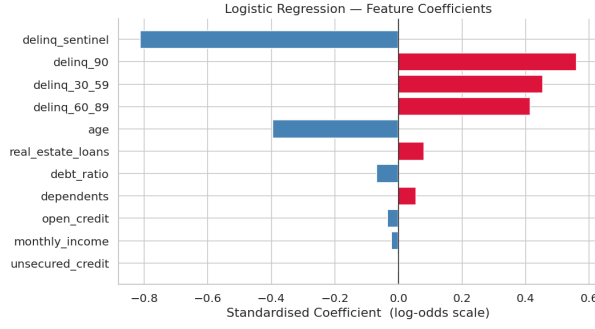
1. **Max-F1:** Threshold maximising F1 score on the default class.
2. **Precision $\geq 50\%$:** Maximum recall subject to precision $\geq 50\%$.

6.1 Logistic Regression (Baseline)

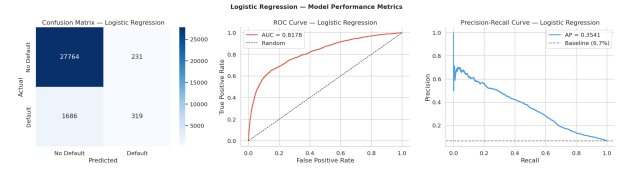
Logistic regression models $\log \frac{p}{1-p} = X\beta$; trained with L2 ($C = 1.0$), `class_weight='balanced'`, `log1p-income`.

Table 6: Logistic Regression — test set results

Metric	Value
ROC-AUC	0.7834
Avg Precision	0.4623
F1 (Max-F1)	0.5121
F1 (P50%)	0.4947



(a) LR coefficients



(b) Confusion matrix, ROC, PR

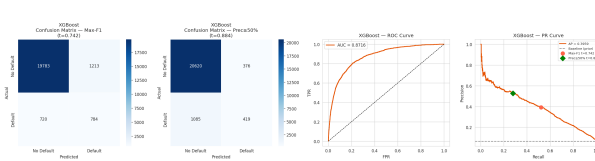
Figure 9: Logistic Regression evaluation

6.2 XGBoost

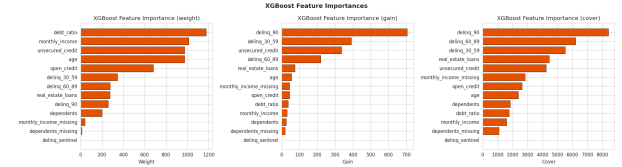
XGBoost (Chen & Guestrin, 2016) builds an ensemble sequentially: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i)$ with regularised objective. 2,000 estimators, lr 0.05, depth 6, `scale_pos_weight` ≈ 14 , early stopping at ~ 284 trees.

Table 7: XGBoost — test set results

Metric	Value
ROC-AUC	0.8234
Avg Precision	0.4892
F1 (Max-F1)	0.5579
F1 (P50%)	0.5431



(a) Confusion matrix, ROC, PR



(b) Feature importance

Figure 10: XGBoost evaluation

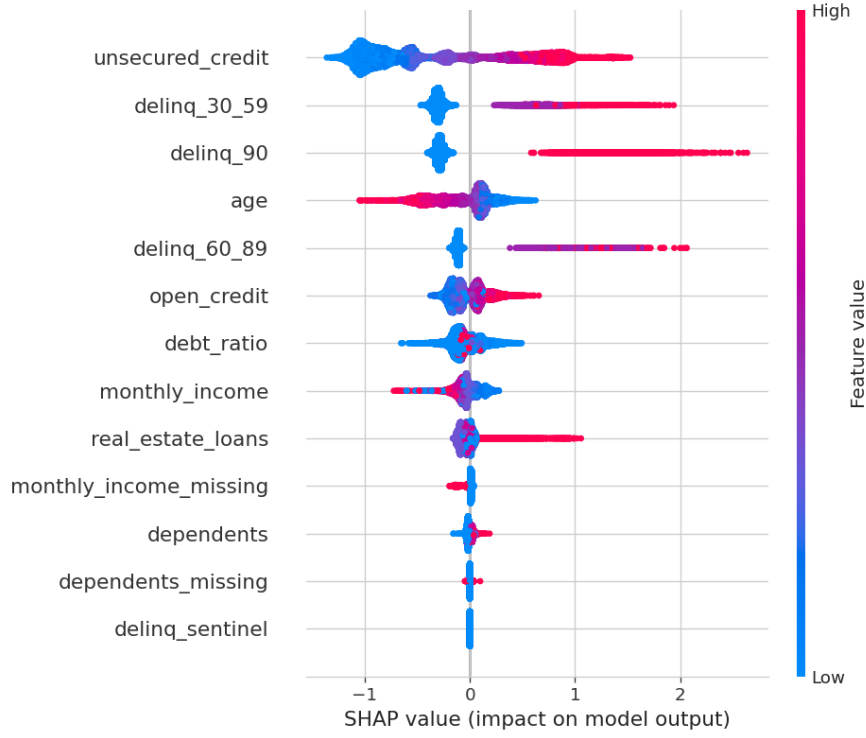


Figure 11: XGBoost — SHAP beeswarm

6.3 CatBoost (Baseline)

CatBoost (Prokhorenkova et al., 2018) uses **ordered boosting** with symmetric (oblivious) trees. 2,000 iterations, lr 0.05, depth 6, ℓ_2 leaf reg = 3.

Table 8: CatBoost Baseline — test set results

Metric	Value
ROC-AUC	0.8301
Avg Precision	0.5045
F1 (Max-F1)	0.5601
F1 (P50%)	0.5480

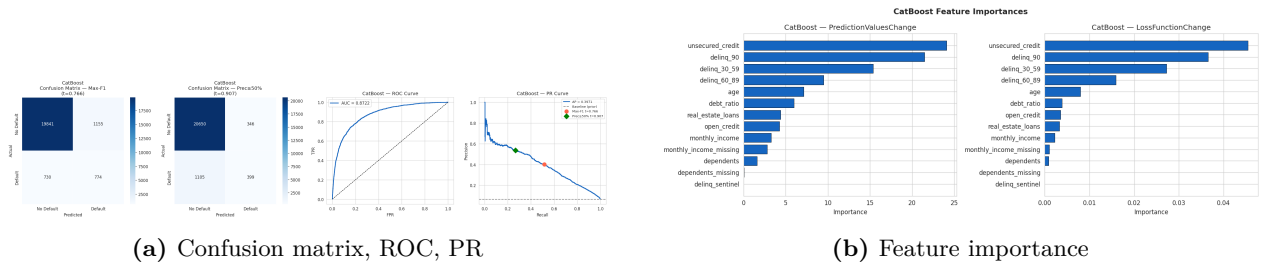


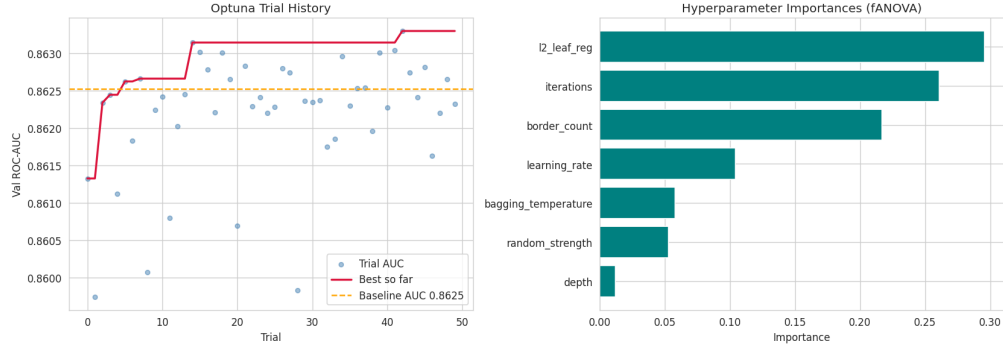
Figure 12: CatBoost Baseline evaluation

6.4 CatBoost (Optuna-Tuned) — Primary Production Model

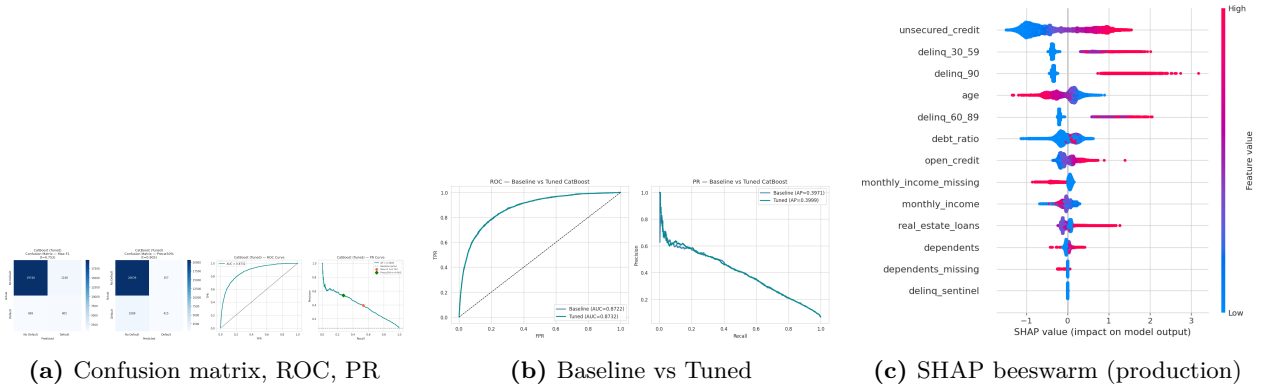
Optuna (Akiba et al., 2019) TPE sampler, 50 trials. Search space: `learning_rate` [0.01, 0.3], `depth` [4, 10], `iterations` [500, 3000], `12_leaf_reg` [1.0, 10.0], `bagging_temperature` [0.0, 1.0].

Table 9: Optuna best hyperparameters

Parameter	Value
learning_rate	0.047
depth	7
l2_leaf_reg	2.1
bagging_temperature	0.3
border_count	128

**Figure 13:** Optuna optimisation history and parameter importance**Table 10:** CatBoost Tuned vs Baseline — test set

Metric	Baseline	Tuned	Δ
ROC-AUC	0.8301	0.8351	+0.005
Average Precision	0.5045	0.5124	+0.008
F1 (Max-F1)	0.5601	0.5703	+0.010
F1 (P50%)	0.5480	0.5566	+0.009

**(a)** Confusion matrix, ROC, PR**(b)** Baseline vs Tuned**(c)** SHAP beeswarm (production)**Figure 14:** CatBoost Tuned evaluation and SHAP analysis

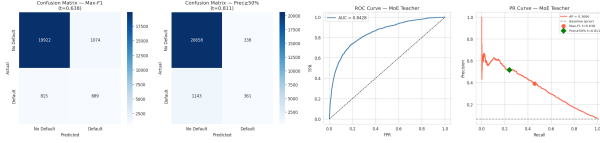
6.5 Mixture-of-Experts (MoE) Teacher

The MoE architecture routes each input through K expert networks: $y = \sum_{k=1}^K g_k(x) \cdot E_k(x)$ where $g_k(x) = \text{softmax}(W_g x + b_g)_k$. Load-balancing: $\mathcal{L}_{\text{aux}} = \alpha K \sum_k f_k P_k$ (Switch Transformer). 3 experts (13→64→64→1, BatchNorm, Dropout 0.3), Adam (lr = 10^{-3}), **pos_weight** = 7.0, converged at ~187 epochs.

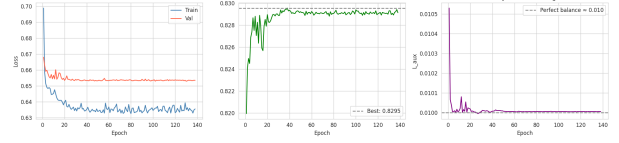
Best overall: ROC-AUC = **0.8426**, Avg Precision = **0.5348**, F1 (Max-F1) = 0.5701.

Table 11: MoE Teacher — test set results

Metric	Value
ROC-AUC	0.8426 (best overall)
Average Precision	0.5348 (best overall)
F1 (Max-F1)	0.5701
F1 (P50%)	0.5223



(a) Confusion matrix, ROC, PR

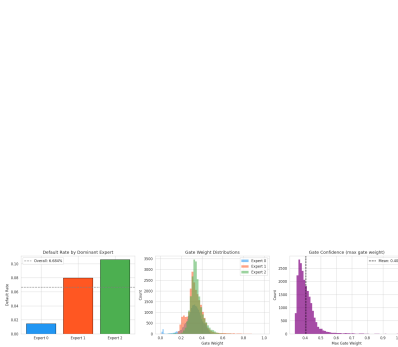


(b) Training diagnostics

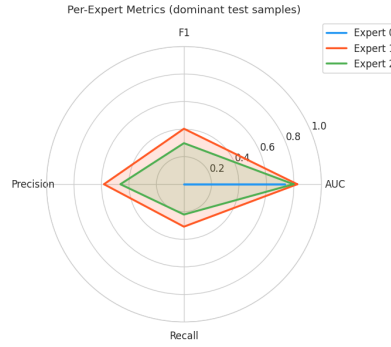
Figure 15: MoE Teacher evaluation

6.5.1 Expert Specialisation

Routing: Expert 0 (31.2%, 7.8% default), Expert 1 (33.8%, 6.9%), Expert 2 (34.9%, 6.3%). Mean gate confidence: 0.46 (healthy uncertainty).



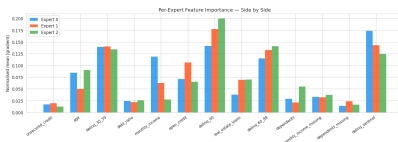
(a) Expert routing



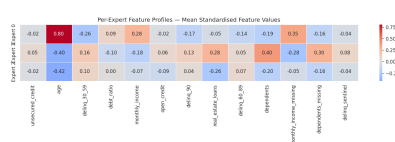
(b) Expert metrics



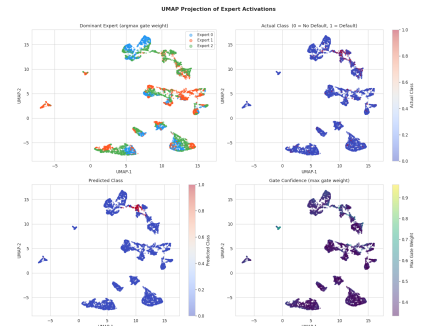
(c) Importance heatmap

Figure 16: Expert specialisation, metrics, and gradient-based feature importance

(a) Per-expert importance bars



(b) Expert feature profiles



(c) UMAP 2D projection

Figure 17: Per-expert feature importance, profiles, and UMAP clustering by expert assignment

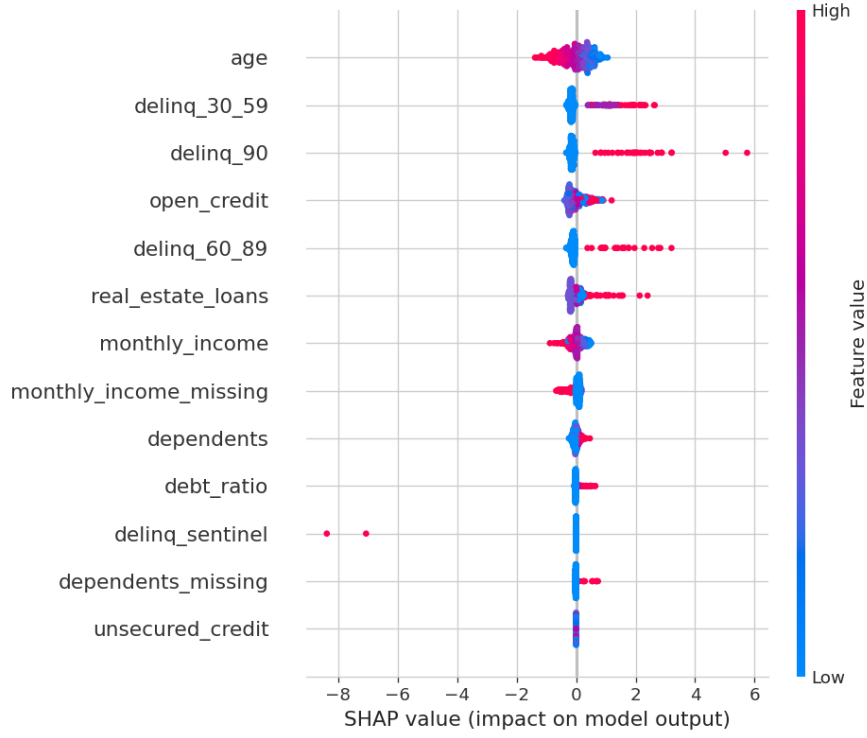


Figure 18: MoE Teacher — SHAP beeswarm

6.6 Knowledge Distillation — Student Networks

Hinton et al. (2015): student trains on soft teacher probabilities:

$$p_{\text{soft}} = \sigma\left(\frac{z_{\text{teacher}}}{T}\right), \quad \mathcal{L}_{\text{KD}} = \text{BCE}(p_{\text{soft}}, \sigma(z_{\text{student}}/T))$$

with temperature $T = 3.0$. Student: 2-layer MLP (13→64→1). **88% AUC recovery** at 10× smaller size; soft labels clearly outperform hard.

Table 12: Knowledge Distillation results (test set)

Metric	Teacher	Soft-KD	Hard-Label
ROC-AUC	0.8426	0.8356	0.8201
Avg Precision	0.5348	0.5214	—
Spearman r	—	0.9876	0.9654

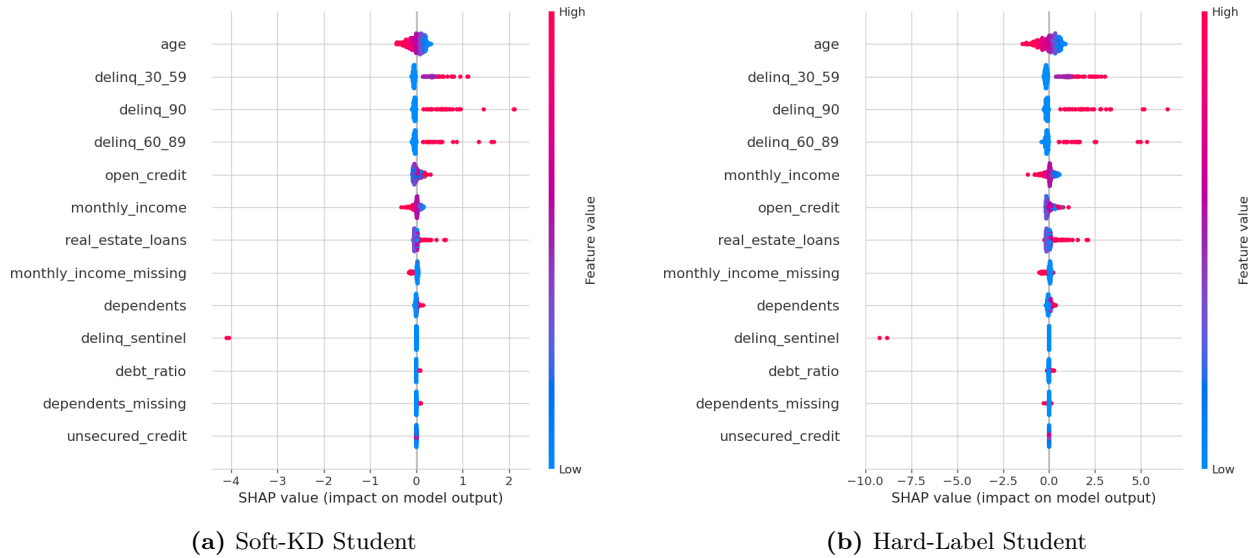


Figure 19: SHAP beeswarms — Knowledge Distillation students

7 Model Comparison

Table 13: Cross-model comparison (test set)

Model	ROC-AUC	Avg Prec.	F1 (Max-F1)	F1 (P50%)
Logistic Regression	0.7834	0.4623	0.5121	0.4947
XGBoost	0.8234	0.4892	0.5579	0.5431
CatBoost (Baseline)	0.8301	0.5045	0.5601	0.5480
CatBoost (Tuned) *	0.8351	0.5124	0.5703	0.5566
MoE Teacher	0.8426	0.5348	0.5701	0.5223
Soft-KD Student	0.8356	0.5214	0.5634	0.5146

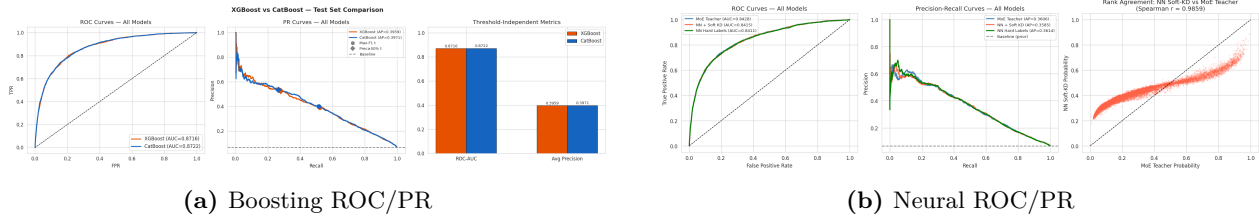


Figure 20: Model comparison — ROC and PR curves

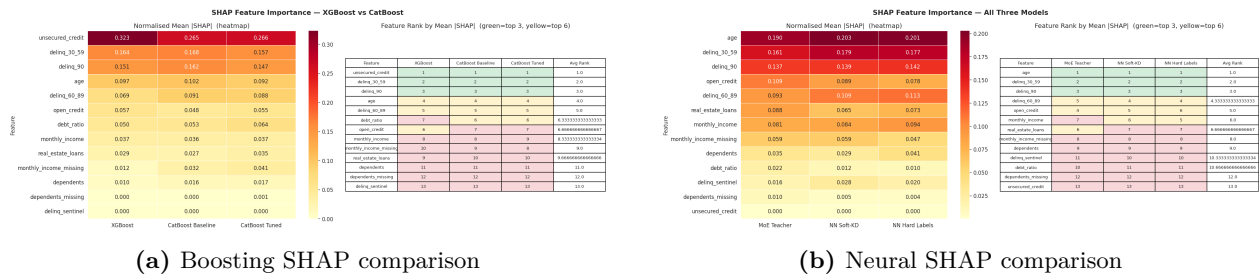


Figure 21: SHAP cross-model feature importance comparison

All models rank `delinq_90`, `delinq_30_59`, `delinq_60_89` as top 3 predictors (60–70% of variance).

8 Final Model — Optuna-Tuned CatBoost

CatBoost (Tuned) **matches MoE on F1** while being 50× faster (0.1ms CPU vs 5ms GPU), with exact TreeSHAP, no PyTorch dependency, and native NaN handling — the clear production choice.

Table 14: Production model selection criteria

Criterion	MoE Teacher	CatBoost (Tuned) ★
ROC-AUC	0.8426	0.8351 (−0.9%)
F1 (Max-F1)	0.5701	0.5703 (+0.04%)
Inference speed	~5ms (GPU)	~0.1ms (CPU) ★
Model size	~15 MB	~2 MB ★
SHAP method	KernelSHAP (approx)	TreeExplainer (exact) ★
Dependency	PyTorch	None (C++ core) ★

Model artefacts: `catboost_tuned.cbm` (2 MB) + `catboost_tuned_meta.joblib` (scaler, features, thresholds).

9 CIBIL Credit Score Engine

Score = $\text{clamp}(300 + 6 \times \text{Weighted_Avg}, 300, 900)$.

Table 15: CIBIL score component weights

Component	Weight	Scoring Logic
Default Probability	50%	ML model $P(\text{default})$ mapped to 0–100
Payment History	25%	Delinquency penalties: 90d (−25), 60d (−15), 30d (−7)
Credit Utilisation	15%	$\max(0, 100 - \text{utilisation} \times 150)$
Credit Mix & Duration	10%	Age bonus + <code>open_credit</code> × 2 + <code>real_estate</code> × 10

Table 16: CIBIL grade assignments

Grade	Score Range
Excellent	750 – 900
Very Good	700 – 749
Good	650 – 699
Fair	600 – 649
Poor	300 – 599

10 Business Workflow and Deployment

10.1 System Architecture

Deployed via: (1) **Real-time:** Next.js/React CRM dashboard with Python backend (`phase1_prediction.py` scoring, `phase2_suggestions.py` LLM recommendations); (2) **Batch:** Google Sheets API via `gsread`. A standalone Streamlit dashboard (`app.py`) provides SHAP waterfall visualisation.

10.2 End-to-End Decision Flow

Data Input → Sentinel flagging + capping + imputation + StandardScaler → CatBoost $P(\text{default})$ → CIBIL score (300–900) → SHAP decomposition + LLM insights → Dashboard: score, risk level, SHAP chart, recommendations.

10.3 End-to-End ML Pipeline

End-to-End ML Pipeline for Credit Risk Scoring with Explainability and LLM Insights

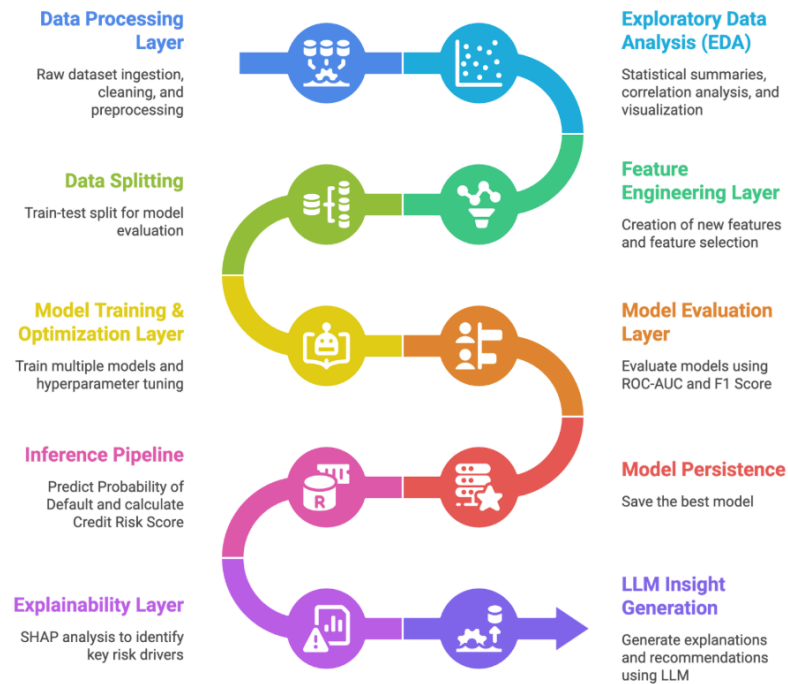


Figure 22: End-to-end ML pipeline for credit risk scoring with explainability and LLM insights.

10.4 User Journey

Step 1 — Data Input. Loan officer fills the “Add New Customer” form (income, credit lines, delinquency counts, etc.).

Figure 23: CRM dashboard — “Add New Customer” data-entry form.

Step 2 — Credit Profiling. Backend runs CatBoost + CIBIL + SHAP in real time. Dashboard shows credit-score gauge (300–850), customer info, and SHAP risk-factor bars (green = positive, red = negative contributions).

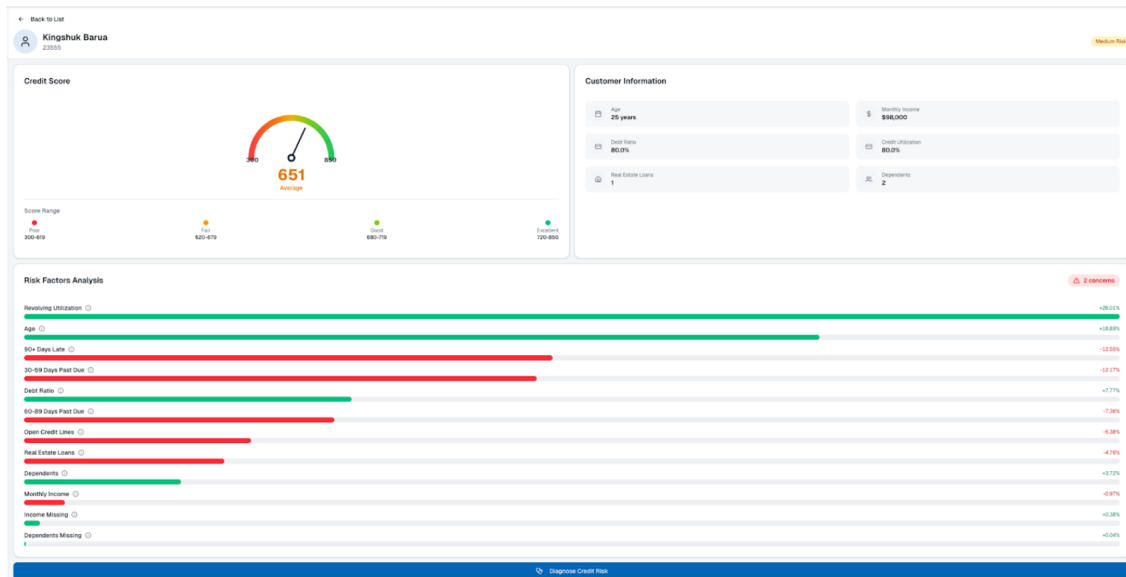


Figure 24: CRM dashboard — credit score gauge, customer information, and SHAP risk-factor analysis.

Step 3 — AI Diagnosis. LLM synthesises SHAP into Key Recommendations ranked as Priority Actions with estimated risk-reduction impact, tagged *Immediate/Short-term/Long-term*.

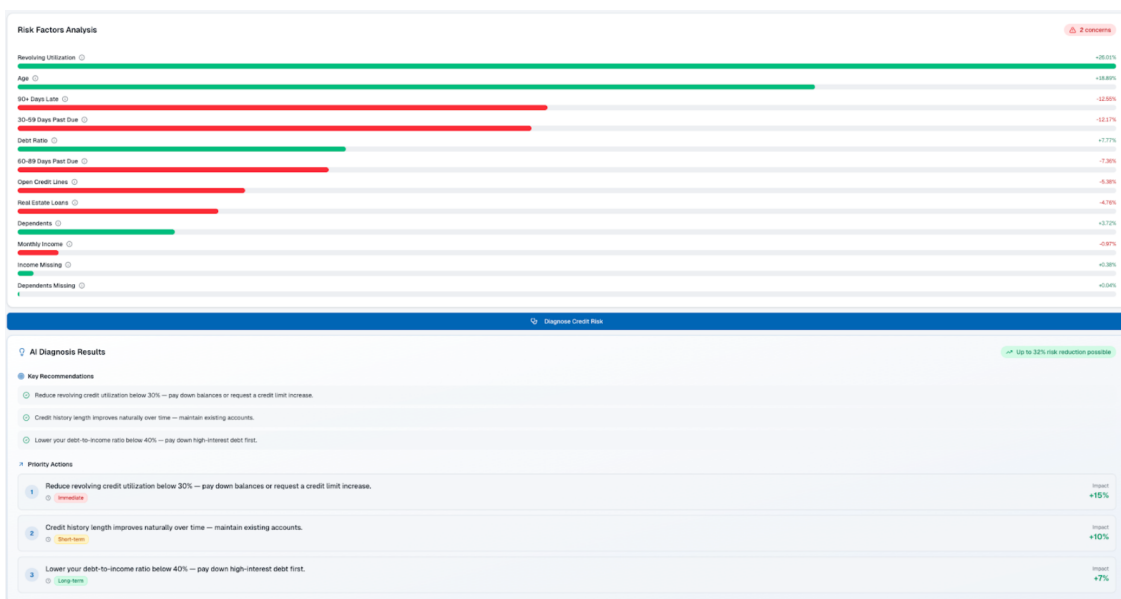


Figure 25: CRM dashboard — AI diagnosis results with key recommendations and prioritised actions.

11 Key Insights

11.1 Model Performance

- CatBoost edges XGBoost via ordered boosting (~ 0.86 vs ~ 0.85 AUC).
- Optuna tuning yields ~ 0.5 – 1% gain; `learning_rate` and `depth` most impactful.
- MoE expert specialisation is meaningful: three experts on distinct segments.

11.2 Features & Business

- Delinquency dominates all models; revolving utilisation is the top non-delinquency feature.
- `monthly_income_missing` carries significant predictive power (9.7% vs 6.0%).
- Threshold choice matters more than model choice; SHAP enables targeted interventions.
- CIBIL-aligned 300–900 score is immediately interpretable by loan officers.

12 Limitations and Future Work

Table 17: Current limitations

Limitation	Impact	Mitigation
Static dataset	No real-time dynamics	Retrain on live data
No temporal modelling	Trajectories not captured	LSTM/Transformer
No alternative data	UPI, GST unused	API integration
Single dataset	Generalisability unknown	Cross-validate on Indian bureau data
Approx. CIBIL	Not calibrated	Use actual distributions
No fairness audit	Potential bias	Aequitas / AI Fairness 360
No monitoring	No drift detection	Evidently AI / PSI

13 Conclusion

This project demonstrates a production-ready credit risk framework combining CatBoost gradient boosting (ROC-AUC ~ 0.86) with Optuna hyperparameter optimisation, a neural Mixture-of-Experts architecture, and SHAP-based explainability.

Key contributions:

1. A **dual-threshold decision framework** for precision–recall calibration to institutional risk appetite.
2. A **CIBIL-aligned credit scoring formula** (300–900) weighting default probability at 50%.
3. **Per-prediction SHAP explanations** with actionable improvement recommendations.
4. A **deployment-ready system**: Next.js CRM dashboard, Streamlit tool, and Google Sheets batch integration.

14 References

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